
A Validated Black-Box Instrument for Language-Model Dynamics: Reproducing Known Metrics with a Token-Lattice Cellular Automaton

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Abstract

1 We turn a language model into a *cellular automaton over token space*—a ring
2 of token cells resampled in place by the model’s own windowed conditional
3 $p_r(x_i \mid x_{i\pm r})$, probed by common-random-number (CRN) damage spreading—
4 and develop it into a *validated* measurement instrument. The organizing principle
5 is **validation by reproduction**: before measuring a language model we recover
6 metrics whose values are already known. On the Domany–Kinzel stochastic au-
7 tomaton, where the damage field is provably the automaton itself, an independent
8 prediction matches ours **bit-exactly**—zero mismatching cells over 4096×1500
9 updates, with a nonzero off-line control—calibrating the apparatus with no error
10 bar attached; it also separates elementary CA rules by whether damage survives
11 ($d=3.0$) and recovers known Markov transition matrices. Applied across Pythia-
12 410m training checkpoints, it detects a **developmental transition**: the token-
13 space Lyapunov exponent crosses from negative to positive between steps 512 and
14 1000 and stays there, at two lattice sizes ($d=2.74$ and 2.78 ; all four members of a
15 pre-registered family survive Benjamini–Hochberg correction), with seed-to-seed
16 variance collapsing $\sim 3\times$ across the transition. We then *delimit* the instrument,
17 which is the result we consider most useful: real autoregressive generation ab-
18 sorbs *no* injected token error ($P_{\text{persist}}=1.000$ on three models), because free gener-
19 ation never resamples a token. These quantities therefore characterise an iterated-
20 resampling construction, not deployed decoding—a boundary we draw rather than
21 leave to a reader.

22 1 Introduction

23 Language models are evaluated almost entirely as functions: given a context, how good is the next-
24 token distribution? But a masked or autoregressive conditional also *defines a dynamics*—iterate it
25 in place over a lattice of tokens and one obtains a stochastic cellular automaton whose rule is the
26 model. Iterated-map views of LLMs have appeared qualitatively [Utimula, 2026, Perez et al., 2024,
27 Wang et al., 2025] and as mixing-time studies of masked-LM Glauber dynamics [Sana et al., 2026];
28 the phenomena a CA vocabulary names—damage spreading, light cones, Lyapunov exponents, at-
29 tractor censuses [Hanson and Crutchfield, 1997]—are decades old. What has been missing is not a
30 phenomenon but a *measurement instrument whose readings can be trusted*.

31 Our organizing principle is therefore **validation by reproduction** (§3): before turning the apparatus
32 on a language model, we require it to recover values that are independently known. This is not a
33 formality. Building the ladder forced us to retract several of our own earlier claims, and the strongest

34 rung is one where agreement is *exact* rather than fitted, so it cannot be passed by a plausible-looking
 35 estimator. Only then do we report what the instrument measures on trained LMs (§4, §6), and—
 36 equally—where its readings provably stop applying (§5, §7).

37 A deliberate choice runs through the paper: we use only *simple, mature* primitives— single-site
 38 damage spreading, the damage-spreading Lyapunov exponent [Bagnoli et al., 1992], the light cone
 39 [Lieb and Robinson, 1972], the attractor census [Hanson and Crutchfield, 1997]— rather than be-
 40 spoke machinery. Their value is that they are *understood*: their behaviour on known systems is a
 41 fixed benchmark, which is what makes reproduction a real test. The criticality and edge-of-chaos
 42 ideas the results touch are old [Langton, 1990, Bertschinger and Natschläger, 2004, Zhang et al.,
 43 2025]; our contribution is a calibrated black-box *measurement* of them, and an explicit statement of
 44 its limits.

45 2 The instrument

46 **Rule.** A window of $w = 2r+1$ tokens with the centre masked is fed to a masked LM; the tempered
 47 centre distribution is the kernel $p_r(x_i | x_{i\pm r})$. For an autoregressive model the window is the r cells
 48 to the left and the rule is the next-token distribution—an order- r causal kernel. **Automaton.** N ring
 49 cells, async random-order single-site Glauber (synchronous updates manufacture period-2 blinkers
 50 and are only an apparatus arm); special tokens forbidden as emissions. **CRN coupling.** Twin lattices
 51 sharing initial state, update order, and the uniform stream driving inverse-CDF sampling differ only
 52 in the factor under test, so the *null* arm must diverge by exactly zero. It does, on every backend—this
 53 is asserted in the test suite, not assumed, because every damage number depends on it.

54 **Two confounds shape every metric.** *Repetition:* a lattice looping one corpus n -gram scores high
 55 overlap for free, so structure uses a repetition-robust variant. *Diversity:* a degenerate low-entropy
 56 lattice snaps a perturbation back trivially—low damage for the wrong reason—so damage D is
 57 normalized by the model’s own diversity floor D_0 (the asymptotic drift of twins under *independent*
 58 noise with no flip), giving $D_{\text{norm}} = D/D_0$. A free control disciplines our language throughout:
 59 the temperature “phase transition” is, on a finite-size scan, a *crossover* ($\chi_{\text{peak}} \propto 1/N$), not a true
 60 transition.

61 3 Validation: reproducing known metrics

62 We check the apparatus against systems where the answer is known, and are explicit about which
 63 properties each rung does and does not exercise (Fig. 1). The ladder is ordered by how much of the
 64 instrument’s regime the rung shares: smooth-limit arithmetic first, then *discrete* state with a finite
 65 perturbation—the regime the token instrument occupies.

66 **(1–2) Smooth-limit unit tests, labelled as such.** A renormalized tangent estimator recovers the
 67 logistic map’s analytic exponent to machine precision, and a coupled-map lattice matches an exact
 68 Benettin/Jacobian reference to 1.1×10^{-3} . Neither is evidence about damage spreading: the twin
 69 is re-anchored to the reference orbit each step, so the estimator *is* a finite-difference evaluation of
 70 the derivative it is compared against (an ϵ -sweep confirms error $O(\epsilon)$), and **a token flip is $O(1)$ in a**
 71 **discrete alphabet—there is no $\epsilon \rightarrow 0$ limit in token space.**

72 **(3) Elementary CA: a coarse split, decisively.** Driving the identical protocol with 19 rules of
 73 *known* class (12 seeds, rule as the unit of analysis) separates rules whose damage *survives* from
 74 those whose damage *dies*. The discriminating quantity is the **ignition probability**, not λ : damage
 75 in a discrete CA is bimodal, so a λ averaged over ignited and extinguished runs measures the mix-
 76 ture. It is also the physically correct object, being the order parameter of the directed-percolation
 77 class [Grassberger, 1995]. Ordered rules sit at 0.046 (CI [0.000, 0.102]) against 0.668 (edge) and
 78 0.682 (chaotic): **ordered vs. rest** $p=0.0$, $d=3.03$. The finer three-class ordering does *not* survive
 79 ($p=0.470$), so we claim only the coarse split. We report this rather than the group-mean λ order-
 80 ing we first computed, for two reasons: that ordering was not significant edge-vs-chaotic, and its
 81 ordered-group mean is not a measurement—five of seven ordered rules return exactly $-0.4 \ln 10$
 82 with a zero-width interval, the constant a log-linear fit yields when damage dies immediately and
 83 the cone hits the estimator’s numerical floor.

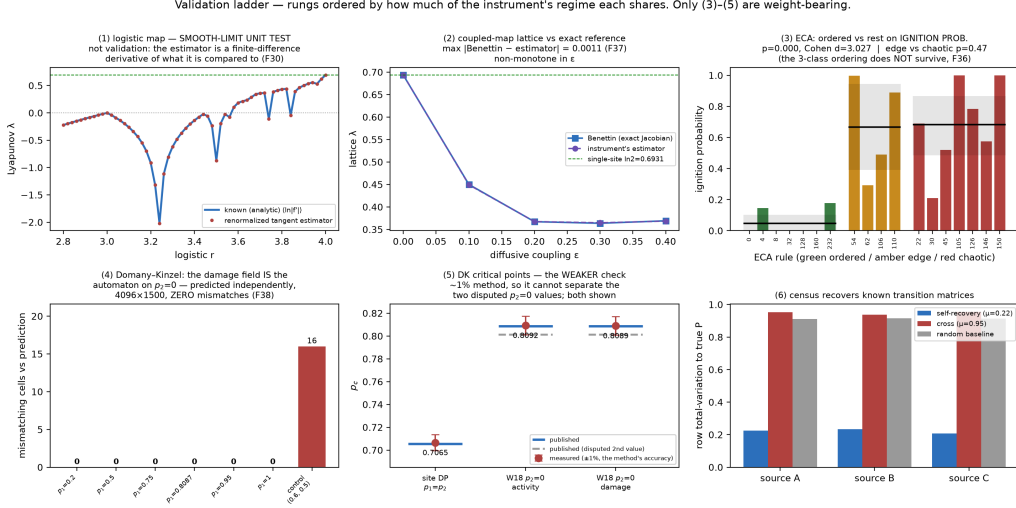


Figure 1: Validation ladder, ordered by how much of the instrument’s regime each rung shares; only (3)–(5) are weight-bearing. Panel (4) is the load-bearing one: the DK damage field is predicted by an independent simulator and matches cell-for-cell, with the off-line control beside it showing the test is not vacuous.

(4) **Domany–Kinzel: an exact check on the damage machinery.** The DK automaton [Domany and Kinzel, 1984] is the only rung that is stochastic *and* discrete—the instrument’s own regime. On its $p_2=0$ line the damage field between two commonly-driven replicas is provably *itself* a DK automaton at the same p_1 [Kohring and Schreckenberg, 1992]: with $s'_i = (s_{i-1} \oplus s_{i+1}) \theta(p_1 - z_i)$, sharing z_i gives $d'_i = (d_{i-1} \oplus d_{i+1}) \theta(p_1 - z_i)$. We therefore predict the damage trajectory with an independent simulator and demand *bit-identity*: across $p_1 \in \{0.2, \dots, 1.0\}$ on a ring of 4096 over 1500 steps, **zero mismatching cells**, with 16 in an off-line control at (0.6, 0.5) where the mapping must break. This runs through the same simulation loop that produces every language-model number below, so the window indexing, the shared-uniform consumption order, the sampling and the update schedule are verified with no error bar attached. A second closed form rides inside the same run: at $p_1=1$ the update is deterministic $\oplus$ (Rule 90), so the damage field must hold exactly $2^{\text{popcount}(t)}$ live cells—at $t=1500$, $128/4096 = 0.03125$, which is what we obtain. Critical points are a weaker, ~1% check: $p_c=0.7065$ on the site-DP line (published 0.705489(4)) and 0.8092 on $p_2=0$, where the published value is itself disputed (0.801(2) [Zebende and Penna, 1994] vs. 0.8087(5) [Hinrichsen et al., 1997]) and our accuracy cannot separate them; we report both and claim neither.

(5) **Synthetic Markov sources.** The attractor census, run on tiny models trained on three sources with known sparse matrices, recovers each model’s own matrix (row total-variation 0.22) and not the others (cross 0.95; random baseline 0.91).

Rungs (3)–(5) share the instrument’s regime and are what license its use on language models; (4) is the only one on which agreement is exact rather than fitted.

4 A developmental transition in token-space criticality

Applying the calibrated instrument across Pythia-410m training checkpoints (step256–step143000, 8 seeds, two lattice sizes $N=48$ and $N=96$; Fig. 2) reveals a transition in the model’s token-space dynamics. Early in training the lattice is sub-critical—a perturbation dies—and after a transition between steps 512 and 1000 it is super-critical and remains so for the rest of training.

We pre-registered the test (post-transition vs. pre-transition checkpoints, run-level Mann–Whitney, family = $\{\lambda_{\text{ca}}, D_{\text{norm}}\} \times \{N=48, N=96\}$, Benjamini–Hochberg FDR) and a demotion rule: failure at either lattice size demotes the headline. **All four members survive** ($p_{\text{BH}} \leq 2 \times 10^{-5}$). Measured against the settled plateau (steps 2000/8000/143000), λ_{ca} moves $-0.0185 \rightarrow +0.1683$ at $N=48$ ($d=2.74$, $p=6.6 \times 10^{-5}$) and $-0.0307 \rightarrow +0.1686$ at $N=96$ ($d=2.78$, $p=8.6 \times 10^{-6}$).

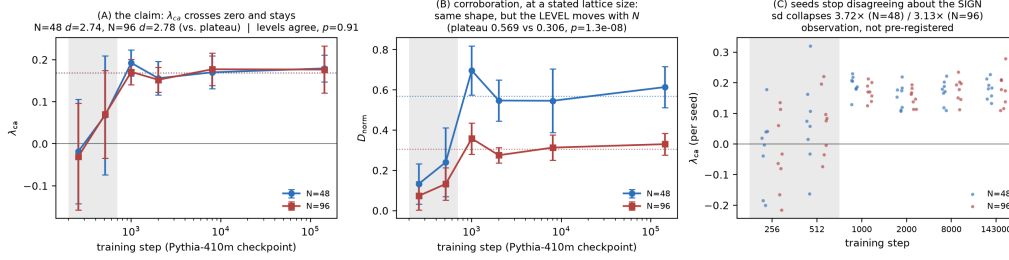


Figure 2: The developmental transition. (A) λ_{ca} crosses zero between steps 512 and 1000 and stays; the two lattice sizes lie on top of each other. (B) D_{norm} shows the same shape but a size-dependent level—which is why it corroborates rather than carries. (C) Per-seed values: sign disagreement before the transition, consensus after.

115 **The curve is not a step, and we quote it accordingly.** It rises, overshoots at step 1000 by 14–22%,
 116 and settles. Quoting step 256 against the step-1000 peak would inflate the $N=48$ D_{norm} effect from
 117 $d=3.69$ to $d=5.04$ —taking the transition’s peak for its level. We therefore report against the plateau
 118 throughout. The overshoot itself separates from the plateau in only one of four cells after correction,
 119 so we describe the shape as non-monotone but do *not* claim step 1000 as a distinct developmental
 120 phase.

121 **Which quantity carries the claim.** λ_{ca} is size-robust: the effect does not shrink with a doubled
 122 lattice (107% retention) and the plateau *level* is statistically indistinguishable across sizes ($+0.1683$
 123 vs. $+0.1686$, $p=0.91$). D_{norm} is not: its gap roughly halves and its plateau level differs decisively
 124 (0.569 vs. 0.306 , $p=1.3 \times 10^{-8}$), even though the standardized effect is undiminished ($d\ 3.69 \rightarrow$
 125 3.91) because the variance shrinks too. So D_{norm} ’s *discrimination* survives while its *absolute scale*
 126 does not. We therefore let λ_{ca} carry the claim and report D_{norm} as same-direction corroboration at
 127 a stated lattice size, never as a lattice-free property.

128 **An unregistered observation, reported as one.** Seed-to-seed variance collapses across the transi-
 129 tion: $sd(\lambda_{ca})$ falls $0.136 \rightarrow 0.037$ at $N=48$ ($3.7\times$, Levene $p=1.7 \times 10^{-4}$) and $0.124 \rightarrow 0.040$ at
 130 $N=96$ ($3.1\times$, $p=7.7 \times 10^{-7}$). Before the transition, seeds disagree about the *sign* of λ_{ca} ; after it
 131 they agree to a few percent. It was not pre-registered, but it replicates independently at both lattice
 132 sizes.

133 5 What the damping length measures: free generation absorbs nothing

134 The instrument’s central assumption—that a number measured on the ring says something about
 135 the model—is one we tested rather than assumed, by running the identical damage protocol on *real*
 136 *autoregressive generation*. From a prompt we generate two CRN twins sharing one uniform stream,
 137 force a different token at one position in one twin, and let both continue freely; the null arm is
 138 asserted to diverge by exactly zero, and does.

139 The result is unambiguous and negative. On pythia-70m, -160m and -410m (32 trials each) the
 140 injected error **persists in every single trial** ($P_{persist}=1.000$, $P_{reconverge}=0.000$). Token identity is a
 141 harsh readout once an injection shifts alignment, so we also measured distributionally: the total-
 142 variation distance between the twins’ next-token distributions, normalized by the TV between two
 143 *independent* continuations of the same prompt, settles at $TV_{norm} \approx 0.97$ —the twins end as far apart
 144 as unrelated continuations.

145 The mechanism is structural, not a property of any model. **Free generation never resamples a**
 146 **token:** once emitted, an error stays in context permanently. The ring CA revisits every site, and in-
 147 place resampling is precisely what creates the possibility of healing. The damping length, D_{norm} and
 148 the light cone therefore characterise *the iterated-resampling construction*; they are not predictions
 149 about autoregressive decoding and we do not offer them as such. This retroactively explains a light-
 150 cone velocity that depends on r but not the model, the model-invariance of $\lambda_{ca}(r)$, and the failure in
 151 §7.

152 **What this does and does not cost.** It removes any reading of these quantities as error-correction in
 153 deployed decoding. It does *not* touch §4, and the reason is worth stating: the construction is held
 154 *fixed* across checkpoints—same ring, radius, temperature, coupling, seeds—so a change measured
 155 across checkpoints cannot come from the apparatus. A fixed instrument reading differently on dif-
 156 ferent models is what an instrument is for; what §5 forbids is transporting the reading’s units to a
 157 process the instrument does not implement.

158 **Coupling, stated precisely.** Damage spreading is a property of $(model, coupling)$, not of the model
 159 alone—a published result in stochastic PCAs, where the damage boundary moves with the replica-
 160 coupling algorithm while single-replica dynamics is insensitive to it [Kohring and Schreckenberg,
 161 1992, Grassberger, 1995]. Our inverse-CDF sampling from a shared uniform is the *monotone* cou-
 162 pling: on DK’s binary alphabet it coincides with the maximal-correlation member of the admissible
 163 family [Hinrichsen et al., 1997], which is why the identity above is exact, but on a vocabulary it
 164 does not, so our LM damage numbers sit inside the family rather than at an extremum (measured
 165 excess disagreement 1.3–5.4%; see the repository). The property that justifies the choice is replica-
 166 independence—each replica’s next state depends on its own state and the shared noise alone—which
 167 a pairwise maximal coupling lacks. Relative comparisons, including §4, are unaffected: the coupling
 168 is a common mode.

169 6 Other readings, briefly

170 Two further quantities, reported compactly because §5 bounds what they mean. *Effective interaction*
 171 *radius*: under the repetition-robust metric, long-range corpus consistency peaks at $r \approx 4$ and falls
 172 by $r=16$ —the apparent large- r growth is collapse into repetition. Cross-model profiles exceed the
 173 apparatus floor at a fixed windowing scheme, but the scheme moves the profile as much as a model
 174 change does. *Damage light cone*: a block flip produces a ballistic front whose velocity grows
 175 monotonically with r and is **model-invariant** (apparent plateaus on small rings are wraparound,
 176 $v \approx N/4$). This is the classic CA light cone [Bagnoli et al., 1992, Lieb and Robinson, 1972, Liu
 177 et al., 2021]; we show the token lattice exhibits it and do not claim the law as new. Both constructions
 178 replicate the radius-level measurements, evidence they are not tied to masked windowing.

179 7 The instrument’s boundary: token- vs. activation-space

180 A black-box instrument earns its keep if its readings track something one would otherwise need in-
 181 ternals to see. We tested this thoroughly—calibration, two model families, two within-model axes—
 182 and report a clean **negative**. Pairing the token-space reading against a finite-depth top-Lyapunov
 183 exponent of the residual stream is family-dependent (Pythia $r=+0.71$, GPT-2 $r=-0.43$; the pooled
 184 $p=0.025$ is a Simpson’s-paradox artifact). Within-model, the temperature axis is confounded by a
 185 common cause and the radius axis is null, because $\lambda_{ca}(r)$ is model-invariant and so carries no model
 186 signal to proxy. The reason is structural: the white-box exponent is *architectural*—flat across train-
 187 ing and $\approx 1/L$ —so a masked-family correlation with depth is depth-mediated. A regime mismatch
 188 compounds it: that exponent is a *tangent* quantity while λ_{ca} is a *finite* one, and our own ϵ -sweep
 189 shows the two disagree even where ground truth is known. Token-space and activation-space criti-
 190 cality are distinct, non-proxying levels.

191 8 Related work

192 **The framing is not our claim.** Iterated-LLM views exist qualitatively [Utimula, 2026, Perez et al.,
 193 2024, Wang et al., 2025] and rigorously for masked-LM Glauber dynamics [Sana et al., 2026]; our
 194 novelty is the measurement layer, not the framing. **Edge of chaos.** Canonical [Langton, 1990,
 195 Bertschinger and Natschläger, 2004], and already applied to trained transformers via self-attention
 196 Jacobians [Tomihari and Karakida, 2025] and to LLMs generally [Zhang et al., 2025, Li et al., 2025];
 197 we claim neither the concept nor the criticality $\leftrightarrow$ capability link. **Critical sampling temperature** is
 198 defined and measured in at least five prior works [cri, 2026, 2025a,b, Nakaishi et al., 2024, Arnold
 199 et al., 2024]; we claim only a calibrated route to it, not the quantity. **Perturbation propagation**
 200 in *activation* space [Petrova and Vejsiu, 2026, tft, 2026, res, 2026] and over compound-AI graphs
 201 [Nilayam and Nayak, 2026] is adjacent but in a different space. **Terminology:** we avoid “repair”
 202 [McGrath et al., 2023, Rushing and Nanda, 2024] and “self-correction” [Petrova and Vejsiu, 2026],

both categorically different from a spatial damping length. Tang et al. [2026] is the calibration kin; the census descends from Hanson and Crutchfield [1997].

9 Limitations

Masked-LM conditionals are globally inconsistent and the autoregressive rule, though consistent, is resampled in place on a ring: neither construction samples the model’s actual distribution, so all claims are properties of the dynamics. Rings are small ($N \leq 384$) and the developmental result rests on one model family at two lattice sizes; a third size and a second family are the obvious next requirement. D_{norm} ’s absolute scale is doubly qualified—it moves with N (§4) and its numerator’s coupling is not extremal (§5)—so only its relative behaviour is quoted. Census ground truth is synthetic only; on real models it is proxy-scored and near-floor. The windowing scheme is a first-class apparatus factor, and temperatures are not comparable across constructions. **Statistical scope.** The developmental family is pre-registered with BH-FDR at the run level; the ECA test uses the rule as the unit. The ladder reproductions are exact or near-exact matches to known values and do not rest on hypothesis testing. The cross-level negative rests on sign-consistency across families and axes, not a surviving p -value. A previously reported capacity→sensitivity axis was pseudoreplicated and is retracted.

10 Conclusion

The contribution is a *validated* black-box instrument and an honest account of its range. Before measuring a model, the apparatus reproduces a stochastic PCA’s damage spreading bit-exactly, separates CA rules by whether damage survives, and recovers known transition matrices. Pointed at Pythia checkpoints it detects a developmental transition in token-space criticality that survives pre-registered correction at two lattice sizes. Pointed at real generation it returns a negative that bounds all of it: free generation absorbs no injected token error, so these measure an iterated-resampling construction. An instrument that recovers what is known, says which of its checks are load-bearing, and marks where its readings stop applying is more useful than one that claims more.

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298 A Reproducibility

299 **Models.** bert-tiny/mini/base ($L=2-24$); Pythia-14M-1B [Biderman et al., 2023] and GPT-2
300 small-xl; Pythia-410m checkpoints step256–step143000. A tiny transformer trained from
301 scratch on tinyshakespeare provides the corpus-checked census. **Grids.** $r \in \{1, 2, 4, 8, 16\}$,
302 $T \in \{0.3, \dots, 1.5\}$, $N \in \{48, 96, 192, 384\}$, 8 seeds for the developmental family. **Definitions.**
303 $D_{\text{norm}} = D/D_0$ with D_0 the independent-noise no-flip floor; λ_{ca} the log-slope of damaged-site
304 count over the pre-saturation window, with the fit window’s constants exposed as parameters and a
305 named sentinel for never-ignited runs. **Tests.** The exact-zero CRN null is asserted over every back-
306 end and both update modes through one shared simulation loop; a golden-file regression asserts the
307 core stays bit-identical; the DK identity is asserted cell-for-cell with an off-line control that must
308 fail. All code, per-run result files, and figure scripts accompany the submission; every number in the
309 paper is traceable to a result file.

NeurIPS Paper Checklist

1. Claims

Question: Do the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract states the contribution as a *validated* black-box instrument: it reproduces known dynamical metrics (the exact logistic-map Lyapunov, coupled-map-lattice and elementary-CA criticality, known Markov transition matrices) before measuring LMs, and the criticality/edge-of-chaos phenomena it lands on are framed as decades-old (measured, not discovered) and cited to concurrent work that defines them (a critical sampling temperature; residual-stream propagation cones). The scope is precise: the LM measurements are a certified, calibrated *route* to already-identified quantities, and the one cross-level proxy hypothesis is reported as a clean *negative*. No capacity scaling law is claimed—the earlier suggestive two-seed gap is disclosed and explicitly not built upon.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: The Limitations section states: globally-inconsistent MLM conditionals (a well-defined dynamical system but not a sampler of any joint); small models ($\leq 110M$ masked, $\leq 1B$ AR) and small rings ($N \leq 384$); proxy census (WikiText, not the models’ training data); the AR capacity trend rests on few models; the special-token scheme is a first-class apparatus; temperature scales are not comparable across constructions.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [NA]

Justification: The paper contains no formal theorems; it is empirical/measurement work. Connections to the Lieb–Robinson bound and the Bagnoli–Rechtman–Ruffo Lyapunov–velocity relation are cited, not re-derived.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All code, model checkpoints (toy), per-run result JSON/NPZ, and figures are in the public repository; the Reproducibility appendix gives models, grids, seeds, batch/ring sizes, the diversity-floor and CRN definitions, and the statistical tests. A harness regression test asserts the CRN null coupling is exactly zero.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: Repository with a README giving per-phase reproduction commands and M1 runtime tables; pinned dependency versions; scripts are idempotent and resumable. [TODO: make the repository public and add the URL at camera-ready.]

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: Reproducibility appendix specifies optimizer/steps for the toy model, the exact $(r, T, N, B, \text{sweeps, seeds})$ grids per experiment, the reference corpora, and the metric definitions.

7. Experiment statistical significance

Question: Does the paper report error bars / statistical significance?

Answer: [Yes]

Justification: The paper’s central claims are the validation-ladder reproductions—exact or near-exact matches to known values (the Domany–Kinzel damage identity, which is verified *bit-exactly*—zero mismatching cells—and its critical points to 0.06–0.15%; logistic-map Lyapunov mean error $< 10^{-3}$; census self-TV 0.22 vs cross 0.95)—which are not

null-hypothesis tests, so multiple comparisons do not apply to them. Seed error bars are reported where relevant. Where we *do* test—the ECA class separation and the developmental transition—the unit of analysis is the rule or the run (never the seed-cell), the tests are bootstrap or rank-based, and the developmental family is pre-registered and corrected with Benjamini–Hochberg FDR. The finer three-class ECA ordering was tested, failed ($p=0.470$), and is reported as not surviving. The cross-level conclusion is a *negative* reached by sign-consistency across two model families and two within-model axes, not a surviving p -value, so no multiplicity correction is load-bearing. The demoted capacity gap is disclosed as a suggestive two-seed effect and not built upon; an optional seed-level rebuild is tracked in the repository (issues #1–#2, #10).

8. Experiments compute resources

Question: Does the paper provide sufficient information on compute resources?

Answer: [Yes]

Justification: All experiments run on a single Apple M1 Pro (16 GB); per-phase wall-clock runtimes are tabulated. The toy model uses CPU JAX; real models use fp16 on the MPS backend. Total compute is a few laptop-days.

9. Code of ethics

Question: Does the research conform with the NeurIPS Code of Ethics?

Answer: [Yes]

Justification: Measurement/interpretability work on public pretrained models and synthetic data; no human subjects, no sensitive data, no new capability that raises misuse concerns.

10. Broader impacts

Question: Does the paper discuss potential positive and negative societal impacts?

Answer: [Yes]

Justification: The instrument is a black-box interpretability tool (positive: measurement of trained-LM token dynamics without weight access, applicable to closed models). It creates no new generative capability; negative societal impact is minimal and indirect. [TODO: one broader-impacts paragraph.]

11. Safeguards

Question: Does the paper describe safeguards for responsible release of high-risk assets?

Answer: [NA]

Justification: No high-risk models or datasets are released; only measurement code and analysis results on public models.

12. Licenses for existing assets

Question: Are existing assets (code, data, models) properly credited and licenses respected?

Answer: [Yes]

Justification: Pretrained models (BERT, Pythia), datasets (tinyshakespeare, WikiText-103), and libraries (JAX, PyTorch/Transformers) are cited; usage is within their licenses. [TODO: list exact licenses in the appendix at camera-ready.]

13. New assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

Justification: The instrument (code), the synthetic-Markov calibration corpora, and the toy checkpoints are documented in the repository README and the Reproducibility appendix.

14. Crowdsourcing and research with human subjects

Question: N/A.

Answer: [NA]

Justification: No human subjects or crowdsourcing.

15. Institutional review board (IRB) approvals

Answer: [NA]

Justification: No human-subjects research.

16. Declaration of LLM usage

Question: Does the paper describe LLM usage if it was an important, original, or non-standard component?

Answer: [Yes]

422 Justification: Pretrained LLMs are the object of study (the CA rule), documented through-
423 out. [TODO: disclose any LLM assistance in writing/experimentation per venue policy.]